

Multiple Choice

1. **Answer:** D

Options: A) Na_2CO_3 ; B) Na_3PO_4 ; C) Na_2S ; D) NaCl

Note: Correct option: D - NaCl

2. **Answer:** C

Options: A) K^+ ; B) Ca^{2+} ; C) Sc^{3+} ; D) Rb^+

Note: Correct option: C - Sc^{3+}

3. **Answer:** D

Options: A) Silicon; B) Diamond; C) Silicon carbide; D) Arsenic-doped silicon

Note: Correct option: D - Arsenic-doped silicon

4. **Answer:** D

Options: A) The process is exothermic.; B) The process does not involve any work.; C) The entropy of the system increases.; D) The total entropy of the system plus surroundings increases.

Note: Correct option: D - The total entropy of the system plus surroundings increases.

5. **Answer:** D

Options: A) $Q = 1012$; B) $Q = 2012$; C) $Q = 3012$; D) $Q = 6012$

Note: Correct option: D - $Q = 6012$

True / False

6. **Answer:** True

Options: A) True; B) False

Note: LLM-generated true statement based on MCQ.

7. **Answer:** True

Options: A) True; B) False

Note: LLM-generated true statement based on MCQ.

8. **Answer:** False

Options: A) True; B) False

Note: LLM-generated false statement. Correct answer: ' SCl_2 '.

9. **Answer:** True

Options: A) True; B) False

Note: LLM-generated true statement based on MCQ.

10. **Answer:** False

Options: A) True; B) False

Note: LLM-generated false statement. Correct answer: 'Ca'.

Long Form Response

11. **Answer:** CH₃Cl. Explain why this is the correct answer and provide supporting evidence. Reference key facts or concepts that support this choice.

Note: Long-form derived from MCQ; award credit for stating the correct idea and giving supporting reasoning.

12. **Answer:** 2-methylpentane. Explain why this is the correct answer and provide supporting evidence. Reference key facts or concepts that support this choice.

Note: Long-form derived from MCQ; award credit for stating the correct idea and giving supporting reasoning.

End of Answer Key